

Multimodal Emotion Recognition Using Speech and Text

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Abstract

This project presents a multimodal emotion recognition system that analyzes emotions using speech, text, and combined multimodal inputs on the TESS dataset. For speech emotion recognition, HuBERT embeddings along with a Bidirectional LSTM (BiLSTM) and attention mechanism were employed to capture temporal emotional characteristics from audio signals.

The text-based pipeline utilized transcribed speech processes through embedding layers and BiLSTM-based contextual modeling. Additionally, a multimodal fusion architecture with cross-attention was implemented to combine speech and text representations effectively.

Experimental results demonstrated that the speech-only model achieved the highest performance, indicating that acoustic features carried significantly stronger emotional information than textual features in the TESS dataset.

A. Architecture Decisions

1. Speech Emotion Recognition Model

Architecture Block	Choice Used	Reason for Using It
Feature Extraction Block	HuBERT	Extracts rich pretrained speech representations more effectively than handcrafted features or MFCCs
Temporal Representation Block	Sequence embeddings ([T x 768])	Preserves temporal emotional variations instead of collapsing sequential information
Sequence Modeling Block	Bidirectional LSTM (BiLSTM)	Captures both past and future speech context for improved emotion understanding
Attention Block	Temporal Attention Soft	Focuses on emotionally important speech frames
Classification Block	Softmax Output Layer	Predicts probability distribution across emotion classes
Optimization Block	Adam Optimizer + CrossEntropyLoss	Provides stable and efficient multi-class training
Overall Architecture	HuBERT + BiLSTM + Attention	Combines pretrained speech representations with temporal emotion modeling

2. Text Emotion Recognition Model

Architecture Block	Choice Used	Reason for Using It
Speech-to-Text Block	HuBERT (HuBERTForCTC)	Automatically converts speech audio into text transcriptions
Text Representation Block	Embedding Layer (Embedding(64))	Converts words into dense semantic vectors
Sequence Modeling Block	Bidirectional LSTM (BiLSTM)	Learns contextual information from both past and future words
Classification Block	Softmax Output Layer	Predicts probability distribution across emotion classes
Optimization Block	Adam Optimizer + Sparse Categorical Crossentropy	Efficient optimization for text classification
Overall Architecture	HuBERT ASR + Embedding + BiLSTM + Dense Classifier	Combines automatic transcription with contextual text emotion modeling

3. Multimodal Fusion Model

Architecture Block	Choice Used	Reason for Using It
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Speech Block	Feature	HuBERT embeddings ([Tx768])	Captures rich acoustic and emotional information
Speech Modeling Block	Sequence	2-layer BiLSTM	Learns temporal emotional patterns from speech
Speech Block	Attention	Attention Pooling	Focuses on emotionally significant speech frames
Text Block	Encoding	BERT (AutoModel)	Extracts contextual semantic representations
Text Representation Block		[CLS] Token Embedding	Provides sentence-level emotional representation
Modality Projection Block		Linear Layers (512→256, 768→256)	Aligns speech and text into a common feature space
Multimodal Fusion Block		Cross Attention (MultiheadAttention)	Learns interactions between speech and text modalities
Modality Weighting Block		Speech = 0.7, Text = 0.3	Assigns higher importance to speech cues
Fusion Block	Strategy	Concatenation Fusion	Combines speech and attended text representations

Classification Block	MLP Classifier	Predicts final emotion class
Optimization Block	AdamW Optimizer	Improves optimization for Transformer-based architectures
Overall Architecture	Multimodal HuBERT + BERT + Cross-Attention Network	Integrates acoustic and semantic emotional understanding

B.Experiments and Comparative Analysis

1. Voice Model

In the TESS dataset, each emotion (7 emotions) contains 200 target words spoken by two different aged women using the carrier phrase:

“Say the word ____”

The emotional distinction mainly comes from the way the sentence is spoken rather than the textual content itself.

The speech-only model achieved the best performance because:

- Emotions are strongly reflected in vocal characteristics
- Pitch, tone, loudness, and speaking style vary significantly across emotions
- Audio signals contain rich emotional patterns

Since each emotion contains 200 samples, the model had sufficient data to learn emotional speech representations effectively.

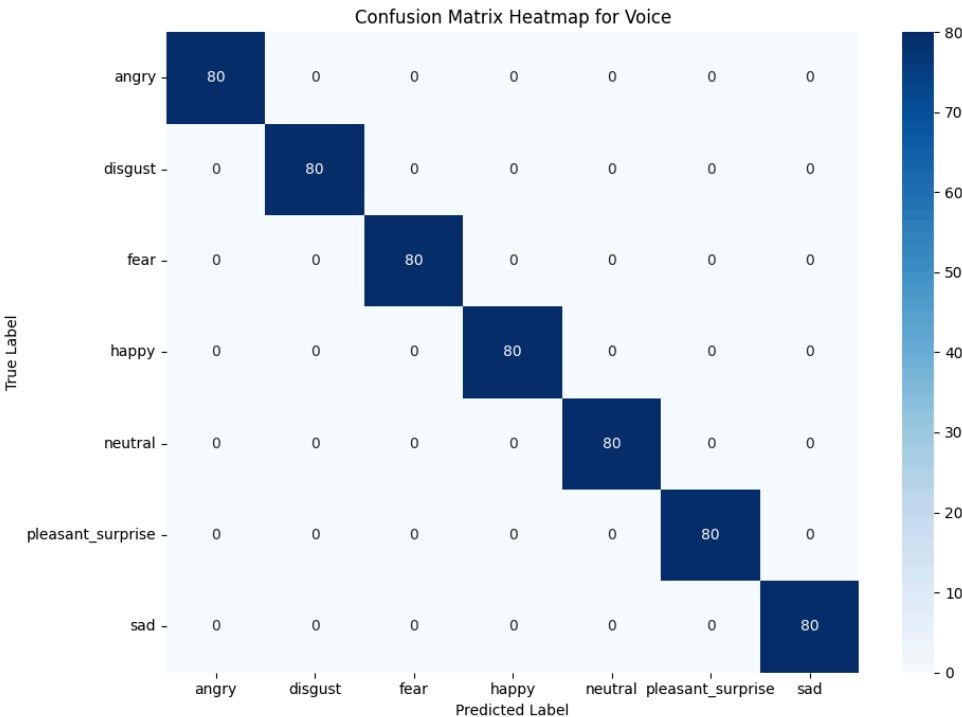


Fig. i. Voice Model Confusion Matrix

The confusion matrix showed strong diagonal dominance, meaning that nearly all samples were correctly classified into their respective emotion categories.

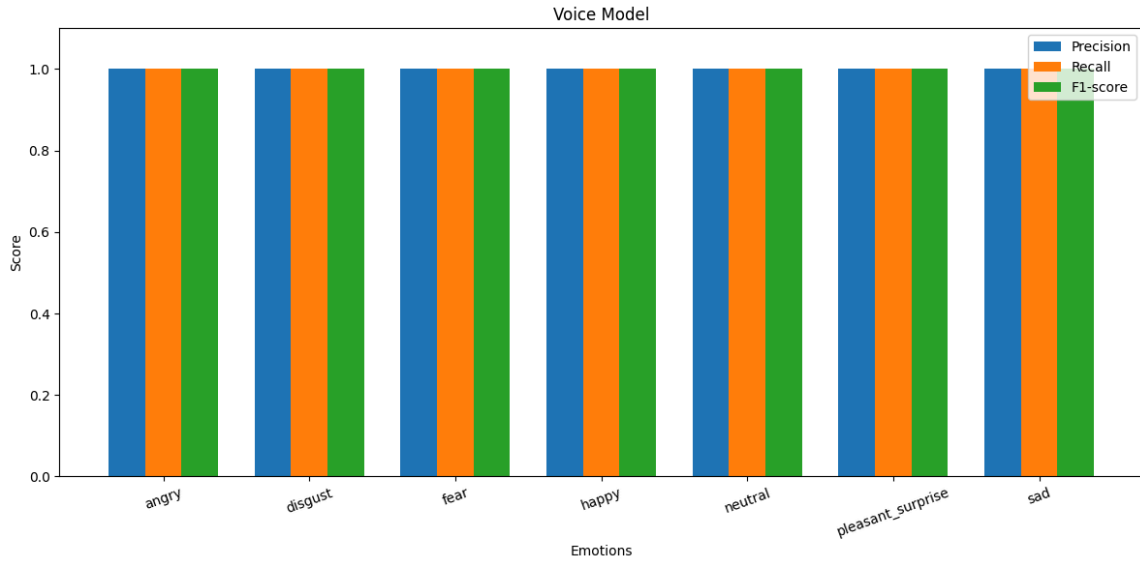


Fig. ii. Voice Model Performance

The speech model achieved nearly perfect classification performance across all seven emotions, with precision, recall, and F1-score values close to 1.0. This indicates that the model classified almost every sample correctly without significant false positives or false negatives.

However, the near-perfect accuracy suggests potential overfitting because the dataset contains only two speakers (OAF and YAF), the model also learns speaker-specific characteristics in addition to emotional patterns.

2. Text Model

The text-only model performed poorly because the dataset contains very limited semantic emotional information. All samples follow the same sentence structure:

“Say the word ____”

The words themselves are random and appear across all emotions. Therefore:

- The text has weak emotional relevance
- The same word can appear in angry, happy, sad, or fear samples
- Emotional meaning is not conveyed through language content

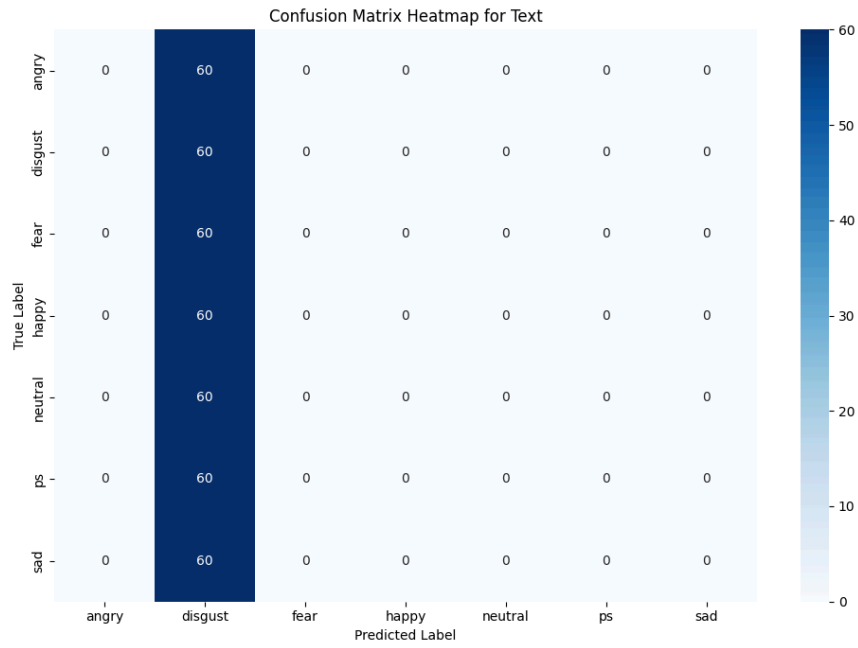


Fig. iii. Text Model Confusion Matrix

The confusion matrix demonstrated that nearly every sample was incorrectly classified as “disgust,” indicating that the text model could not learn useful contextual emotional representations.

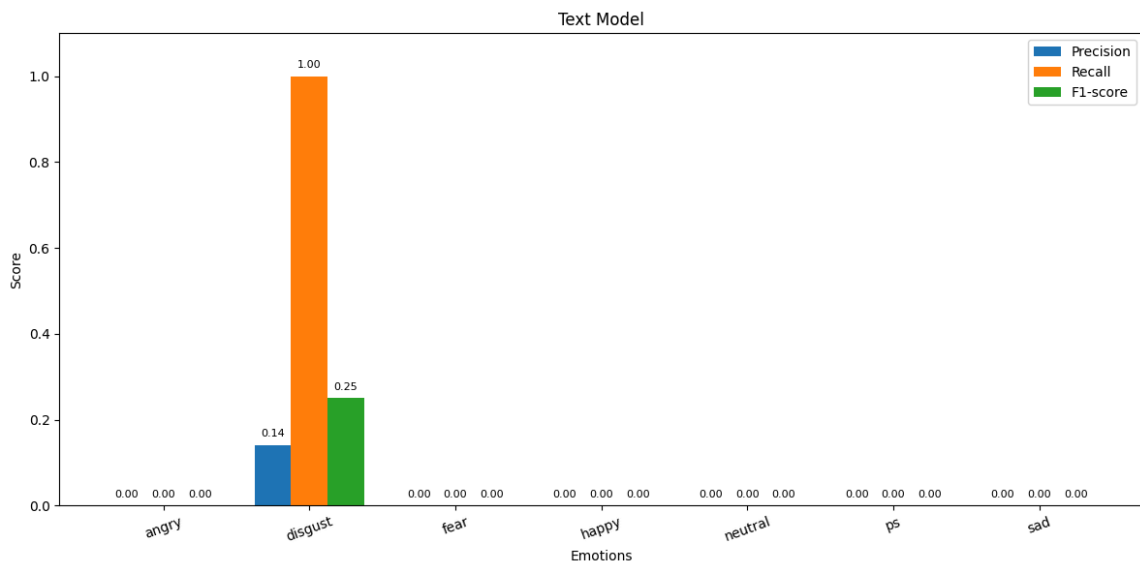


Fig. iv. Text Model Performance

The text model predicted most samples as the “disgust” class, leading to severe class imbalance and poor overall performance.

As a result, the model failed to learn meaningful emotional distinctions and achieved very low classification accuracy.

3. Fusion Model

The multimodal model combined both voice and text representations.

It performed significantly better than the text-only model because speech contributed strong emotional information.

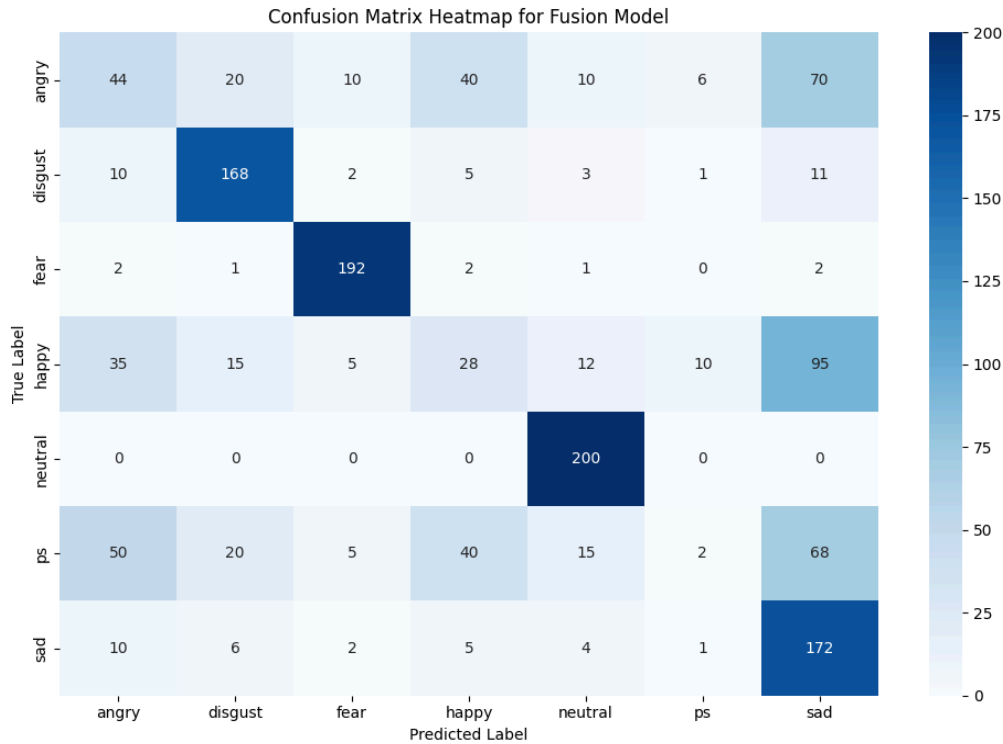


Fig. v. Fusion Model Confusion Matrix

This shows high classification accuracy for neutral, fear, disgust, and sad emotions. Misclassifications mainly occurred between angry, happy, and pleasant surprise because of their similar vocal patterns.

Key observations:

- Speech carries most of the emotional information in TESS
- Text contributes very little semantic value
- Weak text features can negatively affect multimodal fusion performance

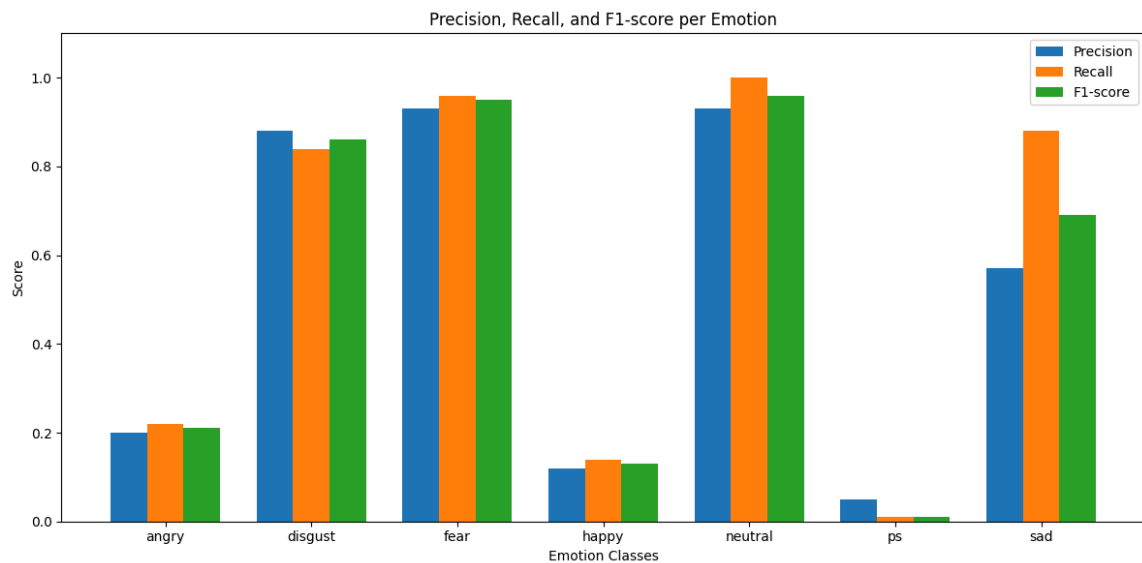


Fig. vi. Fusion Model Performance

The fusion model achieved strong performance for:

- Neutral
- Fear
- Disgust
- Sad

However, emotions such as:

- Angry
- Happy
- Pleasant Surprise (ps)

were more difficult to classify accurately due to overlapping acoustic characteristics.

However, it still underperformed compared to the voice model because the textual modality provided weak and noisy information.

Comparison:

Model	Training Accuracy	Validation Accuracy	Observation
Voice	100%	100%	Achieved the best performance with strong emotional discrimination from vocal cues.

Text	24%	14%	Performed poorly due to limited emotional information in textual content alone.
Fusion	100%	61%	Improved over text-only performance but showed lower generalization than the speech-only model.

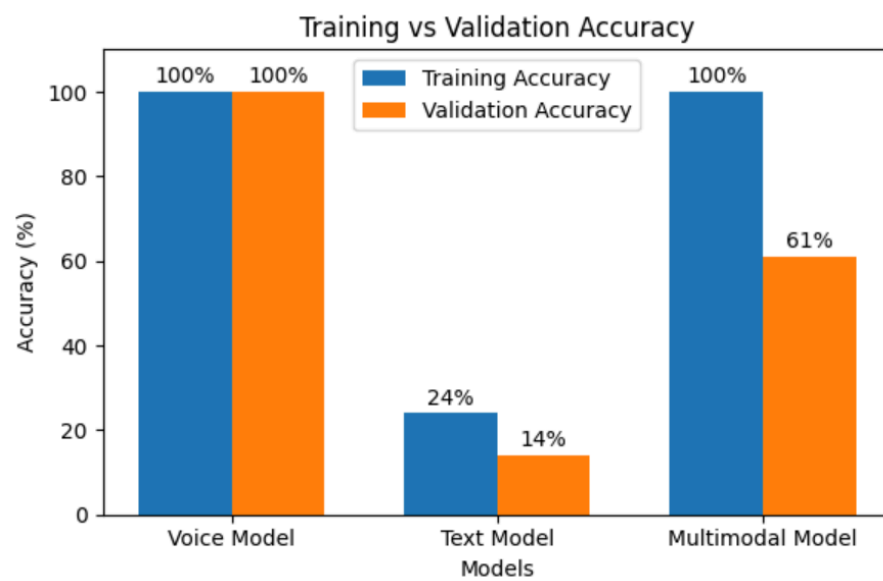


Fig. vii. Comparison of accuracy of models

C. Analysis

1. Easiest and Hardest Emotions to Classify

1.1 Voice Model

Easiest Emotions

- Neutral
- Fear
- Disgust

These emotions exhibited highly distinct acoustic characteristics:

- Neutral - stable tone
- Fear - trembling/high-frequency variations
- Disgust - harsh vocal texture

The HuBERT, BiLSTM, Attention architecture effectively captured these temporal speech variations.

Hardest Emotions

- Happy
- Sad
- Pleasant Surprise
- Angry

These emotions shared overlapping prosodic characteristics:

- Angry and happy both contain high energy
- Sad and pleasant surprise may have similar soft pitch transitions

Speaker dependency also contributed to overlap between emotion clusters as there are only 2 speakers.

1.2 Text Model

The text model struggled because:

- The sentence structure remained fixed
- The words themselves lacked emotional meaning
- Semantic diversity was extremely limited

As a result:

- Most predictions collapsed into the “disgust” class

- Emotion embeddings showed poor separability

1.3 Fusion Model

Easiest Emotions

- Neutral
- Fear
- Disgust
- Sad

Hardest Emotions

- Angry
- Happy
- Pleasant Surprise

These emotions showed strong overlap because:

- Their acoustic patterns were similar
- Text features contributed limited discriminative information

2. When Does Fusion Help Most?

Fusion is most effective when:

- Speech contains ambiguous emotional information
- Text provides complementary context
- Both modalities contribute meaningful emotional cues

In this project, fusion improved performance over the text-only model because audio provided strong emotional signals. However, fusion could not outperform the speech-only model because:

- TESS transcriptions contain weak semantic content
- The sentence structure lacks diversity
- Textual information diluted strong acoustic representations

Fusion is therefore more beneficial in datasets that have contextual importance like:

- Conversational emotion datasets
- Customer support interactions
- Dialogue-based emotion recognition
- Movie and audiovisual emotion datasets

3. Error Analysis

3.1 Speech Model - Failure Cases

1. **Happy as Pleasant Surprise**
Both emotions shared high-energy speech patterns and similar pitch variations.
2. **Angry as Happy**
Both emotions exhibited loud and intense vocal characteristics.
3. **Sad as Neutral**
Some sad samples had low emotional intensity and resembled neutral speech.

3.2 Text Model - Failure Cases

1. No proper classification is done
The words themselves were unrelated to emotional meaning.

3.3 Fusion Model - Failure Cases

1. **Happy as Pleasant Surprise**
Both emotions contained overlapping acoustic characteristics.
2. **Angry as Sad**
Some low-intensity angry samples resembled sad speech.
3. Overlap between Happy, Angry, and Pleasant Surprise
Weak textual information limited fusion effectiveness.

4. Visualization and Emotion Cluster Separability

4.1 Voice

The speech model processes sequential HuBERT embeddings:

[T x 768]

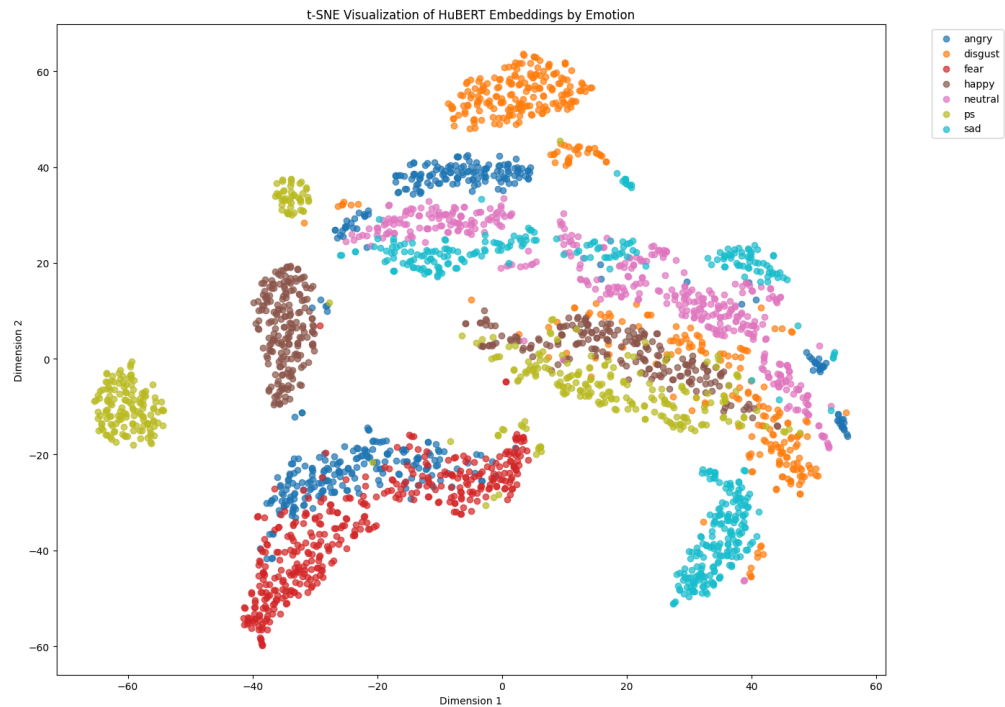


Fig. viii. Voice Model Embedding Distribution

Observations

- Emotion clusters were generally well separated with some overlapping
- Speaker-dependent subclusters were visible
- Significant overlapping

Interpretation

The BiLSTM successfully captured:

- Pitch evolution
- Temporal energy variation
- Speaking rhythm

Attention mechanisms further improved separability by emphasizing emotionally salient speech frames. This can be further improved by making the model not learn the speaker's features but the acoustic features corresponding to the emotions only.

4.2 Text

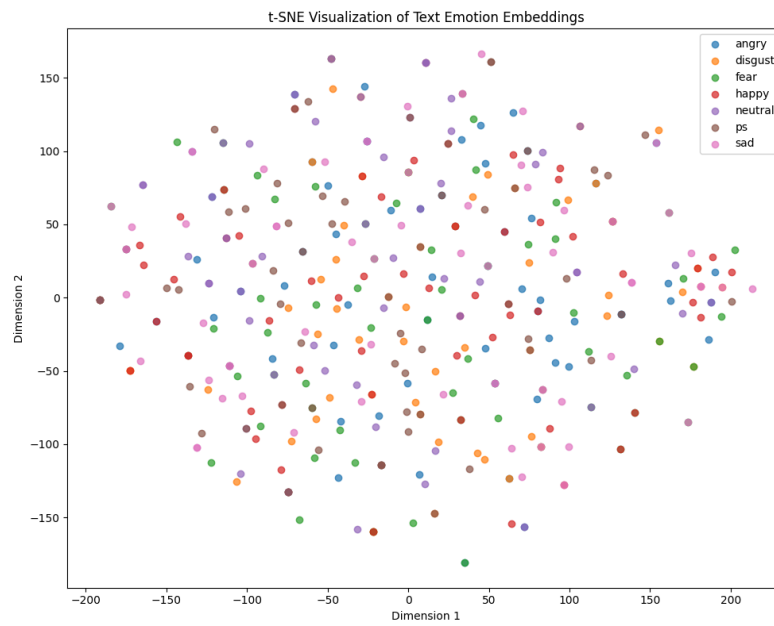


Fig. ix. Text Model Embedding Distribution

Observations

- Emotion clusters were poorly separated
- Embeddings appeared scattered without clear boundaries
- Most samples collapsed toward one dominant prediction region

Interpretation

The lack of semantic emotional diversity prevented meaningful contextual learning.

4.3 Fusion

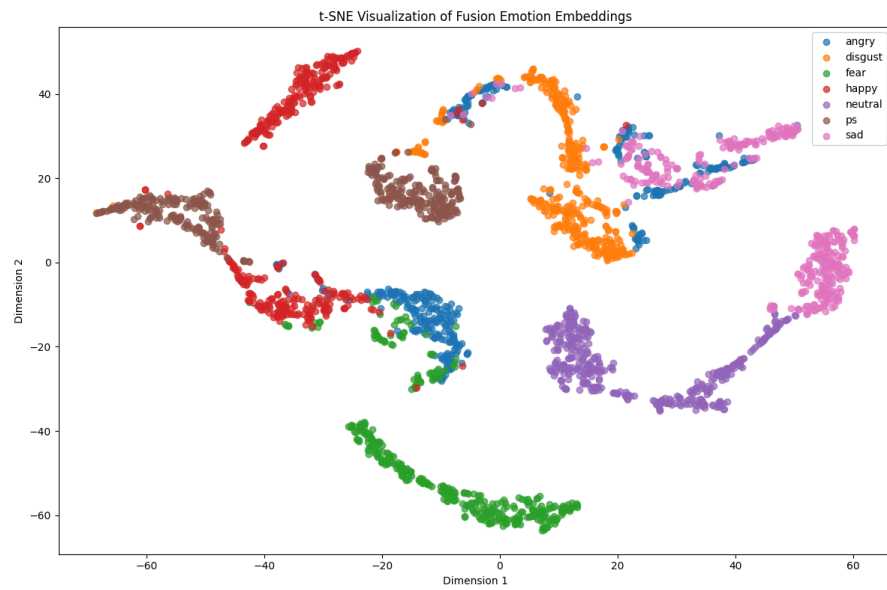


Fig. x. Fusion Model Embedding Distribution

Observations

- Cluster separation improved compared to the text-only model
- Angry, happy, and pleasant surprise still overlapped
- Speaker-based subclusters remained visible

Interpretation

Cross-attention improved interaction between speech and text modalities, enhancing emotional representation robustness. However:

- Weak textual information limited fusion effectiveness
- Speech remained the dominant modality

The fusion embeddings demonstrated that multimodal learning improves robustness only when both modalities contribute meaningful emotional information.

Conclusion

This project demonstrated that speech-based emotion recognition performs better than text and multimodal approaches on the TESS dataset. Acoustic features carried the strongest emotional information, while text contributed very little due to limited semantic context. The study highlights the importance of speech features in emotion recognition tasks.

Github: [Architha-13/Multimodal-Emotion-Recognition](https://github.com/Architha-13/Multimodal-Emotion-Recognition)